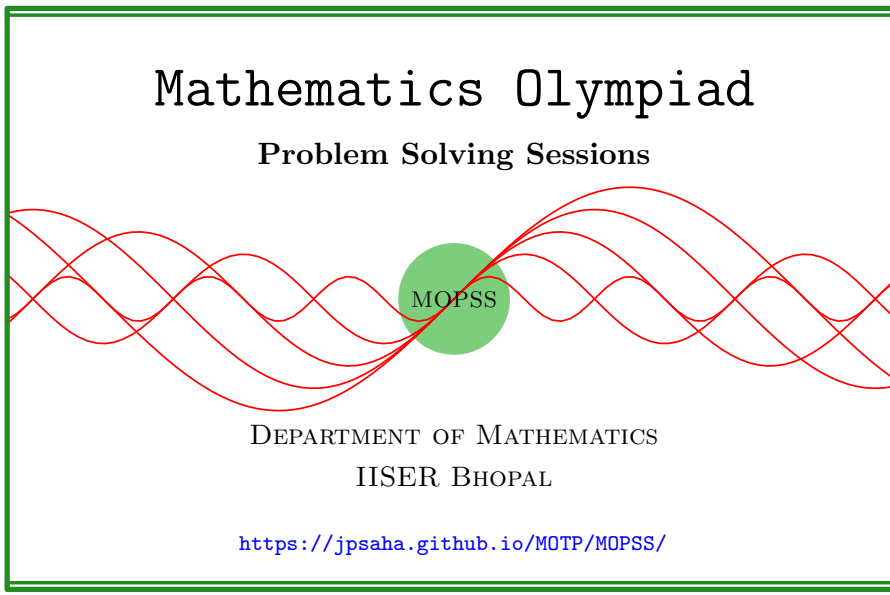
MOPSS

22 November 2025



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**EvanChenOTISExcerpts**].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

Exercise 1 (All-Russian Mathematical Olympiad 2018 Grade 11 Day 1 P2, , by F. Petrov)**.** Let $n \geq 2$ and $x_1, x_2, \ldots, x_n$ be positive real numbers. Prove that

$$\frac{1+x_1^2}{1+x_1x_2} + \frac{1+x_2^2}{1+x_2x_3} + \cdots + \frac{1+x_n^2}{1+x_nx_1} \geq n.$$

Walkthrough —

(a) Use the Cauchy–Schwarz inequality to obtain a lower bound for each term.

(b) Use the AM–GM inequality to complete the proof.

Solution 1. Note that

$$(1+a^2)(1+b^2) \geq (1+ab)^2$$

holds for any real numbers a, b by the Cauchy–Schwarz inequality. This implies that

$$\begin{aligned}
&\frac{1+x_1^2}{1+x_1x_2} + \frac{1+x_2^2}{1+x_2x_3} + \cdots + \frac{1+x_n^2}{1+x_nx_1} \\
&\geq \sqrt{\frac{1+x_1^2}{1+x_2^2}} + \sqrt{\frac{1+x_2^2}{1+x_3^2}} + \cdots + \sqrt{\frac{1+x_n^2}{1+x_1^2}} \\
&\geq n,
\end{aligned}$$

where the last inequality follows from the AM–GM inequality. This completes the proof. ■

Exercise 2 (All-Russian Mathematical Olympiad 2013 Grade 9 Day 1 P1, , by I. Bogdanov)**.** Given three distinct real numbers a, b, and c, show that at least two of the three following equations

$$\begin{aligned}
(x-a)(x-b) &= x-c, \\
(x-b)(x-c) &= x-a, \\
(x-c)(x-a) &= x-b
\end{aligned}$$

have real solutions.

Walkthrough —

(a)

Solution 2. Without loss of generality, let us consider the case that the equations

$$(x-a)(x-b) = x-c, (x-b)(x-c) = x-a$$

do not any real root. It follows that their discriminants are negative, that is,

$$(a+b+1)^2-4(ab+c), (b+c+1)^2-4(bc+a)$$

are negative. Note that

$$\begin{aligned}(a+b+1)^2-4(ab+c) &= (a-b)^2+1+2(a+b-2c)\\ &= (b-a+1)^2+4(a-c),\\ (b+c+1)^2-4(bc+a) &= (b-c)^2+1+2(b+c-2a)\\ &= (b-c+1)^2+4(c-a).\end{aligned}$$

Since a and c are distinct real numbers, one of $a-c$ and $c-a$ is positive. Thus, the discriminant of one of the equations

$$(x-a)(x-b)=x-c, (x-b)(x-c)=x-a$$

is non-negative, which is a contradiction. Therefore, at least two of the three equations have real solutions. ■

Solution 3. Let $f(x)$, $g(x)$, and $h(x)$ denote the polynomials defined by

$$\begin{aligned}f(x) &= (x-a)(x-b)-(x-c),\\ g(x) &= (x-b)(x-c)-(x-a),\\ h(x) &= (x-c)(x-a)-(x-b).\end{aligned}$$

Note that

$$\begin{aligned}f(x)+g(x) &= (x-b)(2x-a-c)-(2x-a-c)\\ &= (2x-a-c)(x-b-1).\end{aligned}$$

This implies that $f(x)+g(x)$ admits a real root. Hence, at least one of $f(x)$ and $g(x)$ is non-positive at some real number. If one of them admits a real root, then we are done. Otherwise, at least one of them is negative at some real number. Also note that if k is a real number satisfying $k \geq |a|+|b|+|c|+2$, then

$$\begin{aligned}f(k+|a|+|b|) &= (k+|a|+|b|+a)(k+|a|+|b|+b)-(k+|a|+|b|+c)\\ &\geq k^2-k-|a|-|b|-|c|\\ &\geq k-|a|-|b|-|c|\\ &> 0\end{aligned}$$

holds, and similarly,

$$g(k+|a|+|b|)>0$$

holds too. By the intermediate value theorem, at least one of $f(x)$ and $g(x)$ admits a real root. Interchanging the roles of f, g, and h, it follows that at least one of any two of the polynomials f, g, and h admits a real root. Therefore, at least two of the three equations have real solutions for x. ■

Exercise 3 (All-Russian Mathematical Olympiad 1999 Grade 9 Day 1 P1, , by S. Volchenkov)**.** The decimal digits of a natural number A form an increasing sequence (from left to right). Find the sum of the digits of $9A$.

> **Walkthrough** —
> **(a)**

Solution 4.

■

Exercise 4 (All-Russian Mathematical Olympiad 2007 Grade 8 Day 1 P1, , by O. Podlipsky)**.** Given reals numbers a, b, c, prove that at least one of three equations

$$\begin{aligned} x^2 + (a-b)x + (b-c) &= 0, \\ x^2 + (b-c)x + (c-a) &= 0, \\ x^2 + (c-a)x + (a-b) &= 0 \end{aligned}$$

has a real root.

> **Walkthrough** —
> **(a)**

Solution 5. Note that the sum of the discriminants of the quadratic polynomials

$$x^2 + (a-b)x + (b-c), x^2 + (b-c)x + (c-a), x^2 + (c-a)x + (a-b)$$

are equal to

$$\begin{aligned} &(a-b)^2 - 4(b-c) + (b-c)^2 - 4(c-a) + (c-a)^2 - 4(a-b) \\ &= (a-b)^2 + (b-c)^2 + (c-a)^2, \end{aligned}$$

which is non-negative, and hence, at least one of the discriminants is non-negative. This implies that at least one of the three equations has a real root. ■

Solution 6. Consider the three quadratic polynomials defined by

$$\begin{aligned} f(x) &= x^2 + (a-b)x + (b-c), \\ g(x) &= x^2 + (b-c)x + (c-a), \\ h(x) &= x^2 + (c-a)x + (a-b). \end{aligned}$$

Note that

$$f(0) + g(0) + h(0) = 0,$$

which implies that at least one of $f(0)$, $g(0)$, $h(0)$ is non-positive. Without loss of generality, assume that $f(0) \leq 0$. If $f(0)$ is zero, then $x = 0$ is a root of $f(x)$. If $f(0) < 0$, then noting that

$$\begin{aligned}
&f(2|a-b| + 2|b-c| + 1) \\
&= \frac{1}{2}(2|a-b| + 2|b-c| + 1)^2 + (a-b)(2|a-b| + 2|b-c| + 1) \\
&\quad + \frac{1}{2}(2|a-b| + 2|b-c| + 1)^2 + (b-c) \\
&= \frac{1}{2}(2|a-b| + 2|b-c| + 1)\,(2|a-b| + 2|b-c| + 1 - 2(a-b)) \\
&\quad + \frac{1}{2}(2|a-b| + 2|b-c| + 1)^2 + (b-c) \\
&\geq 1
\end{aligned}$$

holds, it follows from the intermediate value theorem that $f(x)$ has a real root. ■